

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

11

10 answers · Rationale-rich · Teach from doc

Q1**120**

The correct answer is 120. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ can be calculated using the quadratic formula and are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -8$, and $c = -7$. It follows that the solutions to the given equation are $x = \frac{8 \pm \sqrt{8^2 - 4(2)(-7)}}{2(2)}$, which is equivalent to $x = \frac{8 \pm \sqrt{64 + 56}}{4}$, or $x = \frac{8 \pm \sqrt{120}}{4}$. It's given that one solution to the equation $2x^2 - 8x - 7 = 0$ can be written as $\frac{8 - \sqrt{k}}{4}$. The solution $\frac{8 - \sqrt{120}}{4}$ is in this form. Therefore, the value of k is 120.

Q2**A**A. $\frac{59}{5}$

Choice A is correct. The value of k for which $fk = 3k$ can be found by substituting k for x and $3k$ for fx in the given equation, $fx = 59 - 2x$, which yields $3k = 59 - 2k$. For this equation to be true, either $-3k = 59 - 2k$ or $3k = 59 - 2k$. Adding $2k$ to both sides of the equation $-3k = 59 - 2k$ yields $-k = 59$. Dividing both sides of this equation by -1 yields $k = -59$. To check whether -59 is the value of k , substituting -59 for k in the equation $3k = 59 - 2k$ yields $3(-59) = 59 - 2(-59)$, which is equivalent to $-177 = 177$, or $-177 = 177$, which isn't a true statement. Therefore, -59 isn't the value of k . Adding $2k$ to both sides of the equation $3k = 59 - 2k$ yields $5k = 59$. Dividing both sides of this equation by 5 yields $k = \frac{59}{5}$. To check whether $\frac{59}{5}$ is the value of k , substituting $\frac{59}{5}$ for k in the equation $3k = 59 - 2k$ yields $3\left(\frac{59}{5}\right) = 59 - 2\left(\frac{59}{5}\right)$, which is equivalent to $\frac{177}{5} = \frac{177}{5}$, or $\frac{177}{5} = \frac{177}{5}$, which is a true statement. Therefore, the value of k for which $fk = 3k$ is $\frac{59}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**50**

The correct answer is 50. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. Applying the distributive property to the left-hand side of the equation $kx^2 - 56x - 16 = 0$ yields $kx^2 - 56x - 16 = 0$. Adding 16 to each side of this equation yields $kx^2 - 56x + 16 = 0$. Substituting k for a , -56 for b , and 16 for c in $b^2 - 4ac$ yields a discriminant of $-56^2 - 4k(16)$, or $3,136 - 64k$. If the given equation has no real solution, it follows that the value of $3,136 - 64k$ must be negative. Therefore, $3,136 - 64k < 0$. Adding $64k$ to both sides of this inequality yields $3,136 < 64k$. Dividing both sides of this inequality by 64 yields $49 < k$, or $k > 49$. Since it's given that k is an integer, the least possible value of k is 50.

Q4**A****A. -6**

Choice A is correct. It's given that the graph of $y = fx$ in the xy -plane passes through the points $7, 0$ and $-3, 0$. It follows that when the value of x is either 7 or -3 , the value of fx is 0 . It's also given that the function f is defined by $fx = ax^2 + bx + c$, where a , b , and c are constants. It follows that the function f is a quadratic function and, therefore, may be written in factored form as $fx = a(x - u)(x - v)$, where the value of fx is 0 when x is either u or v . Since the value of fx is 0 when the value of x is either 7 or -3 , and the value of fx is 0 when the value of x is either u or v , it follows that u and v are equal to 7 and -3 . Substituting 7 for u and -3 for v in the equation $fx = a(x - u)(x - v)$ yields $fx = a(x - 7)(x + 3)$, or $fx = ax^2 - 4ax - 21a$. Distributing the right-hand side of this equation yields $fx = ax^2 - 7ax + 3ax - 21a$, or $fx = ax^2 - 4ax - 21a$. Since it's given that $fx = ax^2 + bx + c$, it follows that $b = -4a$. Adding a to each side of this equation yields $a + b = -3a$. Since $a + b = -3a$, if a is an integer, the value of $a + b$ must be a multiple of 3 . If a is an integer greater than 1 , it follows that $a \geq 2$. Therefore, $-3a \leq -6$. It follows that the value of $a + b$ is less than or equal to -6 . Of the given choices, only -6 is a multiple of 3 that's less than or equal to -6 .

Choice B is incorrect. This is the value of $a + b$ if a is equal to, not greater than, 1 .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**5**

The correct answer is 5. Dividing each side of the given equation by 3 yields $x^2 - 6x - 5 = 0$. Adding 5 to each side of this equation yields $x^2 - 6x = 5$. Therefore, if $3x^2 - 18x - 15 = 0$, the value of $x^2 - 6x$ is 5.

Q6**D**

D. 11.00

Choice D is correct. The value of $p(12)$ depends on the value of the constant c , so the value of c must first be determined. It is given that $p(c) = 10$. Based on the definition of p , it follows that:

$$p(c) = \frac{(c - c)^2 + 160}{2c} = 10$$

$$\frac{160}{2c} = 10$$

$$2c = 16$$

$$c = 8$$

This means that $p(x) = \frac{(x - 8)^2 + 160}{16}$ for all values of x . Therefore:

$$p(12) = \frac{(12 - 8)^2 + 160}{16}$$

$$= \frac{16 + 160}{16}$$

$$= 11$$

Choice A is incorrect. It is the value of $p(8)$, not $p(12)$. Choices B and C are incorrect. If one of these values were correct, then $x = 12$ and the selected value of $p(12)$ could be substituted into the equation to solve for c . However, the values of c that result from choices B and C each result in $p(c) < 10$.

Q7**14.5**

Accept also: 29/2

The correct answer is $\frac{29}{2}$. According to the first equation in the given system, the value of y is -1.5 . Substituting -1.5 for y in the second equation in the given system yields $-1.5 = x^2 + 8x + a$. Adding 1.5 to both sides of this equation yields $0 = x^2 + 8x + a + 1.5$. If the given system has exactly one distinct real solution, it follows that $0 = x^2 + 8x + a + 1.5$ has exactly one distinct real solution. A quadratic equation in the form $0 = px^2 + qx + r$, where p , q , and r are constants, has exactly one distinct real solution if and only if the discriminant, $q^2 - 4pr$, is equal to 0 . The equation $0 = x^2 + 8x + a + 1.5$ is in this form, where $p = 1$, $q = 8$, and $r = a + 1.5$. Therefore, the discriminant of the equation $0 = x^2 + 8x + a + 1.5$ is $8^2 - 4(1)(a + 1.5)$, or $64 - 4a - 6$, or $58 - 4a$. Setting the discriminant equal to 0 to solve for a yields $58 - 4a = 0$. Adding $4a$ to both sides of this equation yields $58 = 4a$. Dividing both sides of this equation by 4 yields $\frac{58}{4} = a$, or $\frac{29}{2} = a$. Therefore, if the given system of equations has exactly one distinct real solution, the value of a is $\frac{29}{2}$. Note that $29/2$ and 14.5 are examples of ways to enter a correct answer.

Q8**A****A.** $V(x) = 11x(x - 1)$

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is 1 inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is 11 inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**D**

D. $t = \frac{2u}{u-3}$

Choice D is correct. Multiplying both sides of the given equation by $t-2$ yields $(t-2)(u-3) = 6$. Dividing both sides of this equation by $u-3$ yields $t-2 = \frac{6}{u-3}$. Adding 2

to both sides of this equation yields $t = \frac{6}{u-3} + 2$, which can be rewritten as

$t = \frac{6}{u-3} + \frac{2(u-3)}{u-3}$. Since the fractions on the right-hand side of this equation have a

common denominator, adding the fractions yields $t = \frac{6+2(u-3)}{u-3}$. Applying the distributive

property to the numerator on the right-hand side of this equation yields $t = \frac{6+2u-6}{u-3}$, which

is equivalent to $t = \frac{2u}{u-3}$.

Choices A, B, and C are incorrect and may result from various misconceptions or miscalculations.

Q10

D

D. $y = 2004^x$

Choice D is correct. Since the value of y increases by a constant factor, 4, for each increase of 1 in the value of x , the relationship between x and y is exponential. An exponential relationship between x and y can be represented by an equation of the form $y = ab^x$, where a is the value of y when $x = 0$ and y increases by a factor of b for each increase of 1 in the value of x . Since $y = 200$ when $x = 0$, $a = 200$. Since y increases by a factor of 4 for each increase of 1 in the value of x , $b = 4$. Substituting 200 for a and 4 for b in the equation $y = ab^x$ yields $y = 2004^x$. Thus, the equation $y = 2004^x$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This equation represents a relationship where for each increase of 1 in the value of x , the value of y increases by a factor of 200, not 4, and when $x = 0$, y is equal to 4, not 200.

Choice C is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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